

Hannah Cyberey

📍 Charlottesville, VA ✉ yc4dx@virginia.edu 🔗 hannahxchen.github.io 🌐 hannahxchen

Education

Ph.D. in Computer Science 2019 – 2025

University of Virginia, Charlottesville, VA

- Dissertation: *Sensitivity Auditing for Trustworthy Language Models*
- UVA Engineering Dean's Scholar Fellowship (2019 – 2024)

B.S. in Information Management 2014 – 2018

Chang Gung University, Taiwan

- Presidential Awards (Top 3% of class): 2016 Fall, 2017 Spring, and 2017 Fall

Experience

Postdoctoral Research Associate Sep 2025 – Present

School of Data Science, University of Virginia | Charlottesville, VA

- Led statistical analysis for a study of how AI research translates into open-source infrastructure, fitting matched case-control and survival models with cluster-robust inference to identify what predicts integration of papers into HuggingFace libraries
- Built pipelines linking papers, code, and contributors across scholarly databases and GitHub; validated automated paper-to-code links against manually coded samples, estimating precision for each link source
- Developed a white-box auditing framework for LLMs that operationalizes abstract model properties as measurable constructs and surfaces risks standard evaluations miss (TMLR 2026)
- Advised PhD students on benchmark alignment project, calibrating LLM benchmarks to reliably predict preference rankings for unseen frontier models

Graduate Research Assistant Apr 2019 – Aug 2025

Department of Computer Science, University of Virginia | Charlottesville, VA

- Developed unsupervised methods for extracting steering vectors from model activations to detect and control behaviors such as censorship, refusal, and bias at inference time (COLM 2025, EMNLP 2025)
- Evaluated the construct validity of prevalent LLM bias metrics in resume screening and essay grading tasks, demonstrating that the metrics fail to capture allocational harms that arise when model predictions are used to allocate limited resources (Insights Workshop 2025)
- Analyzed gender bias in NLP models under statistical and causal fairness notions, showing that debiasing methods targeting one notion can worsen the other (NAACL Findings 2024)
- Designed adversarial evaluations and test cases for NLP models, developing balanced adversarial training to address oversensitivity and undersensitivity tradeoffs under attack (EMNLP 2022)
- Deployed a public interactive demo of the censorship steering research on HuggingFace Spaces, with a custom Docker/FastAPI model backend and a Gradio interface for configurable steering
- Published 5 peer-reviewed first-author papers at top NLP venues; mentored 5 undergraduate students on independent research projects

Research Intern Feb 2022 – May 2022

Microsoft Research | Redmond, WA

- Investigated privacy leakage and copyright risks in code generation models within a cross-functional team spanning research, legal, and engineering
- Conducted threat modeling to assess privacy risks via membership inference and training data extraction attacks; proposed mitigation methods to improve the deployment pipeline

- Presented research findings to senior leadership, translating technical results into actionable guidance

Machine Learning Intern

Jun 2018 – Dec 2018

Cybersecurity Technology Institute, Institute for Information Industry | Taiwan

- Analyzed security vulnerability trends on Twitter and built binary classifiers to detect threats at scale
- Developed Sec2Vec, a domain-specific embedding method to improve security threat detection accuracy

Publications

Hannah Cyberey, Yangfeng Ji, David Evans. [White-Box Sensitivity Auditing with Steering Vectors](#). In *Transactions on Machine Learning Research (TMLR)*, Jun 2026.

Hannah Cyberey, Yangfeng Ji, David Evans. [Unsupervised Concept Vector Extraction for Bias Control in LLMs](#). In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Nov 2025.

Hannah Cyberey, David Evans. [Steering the CensorShip: Uncovering Representation Vectors for LLM “Thought” Control](#). In *Conference on Language Modeling (COLM)*. Oct 2025.

Hannah Cyberey, Yangfeng Ji, David Evans. [Do Prevalent Bias Metrics Capture Allocational Harms from LLMs?](#) In *Proceedings of the Fifth Workshop on Insights from Negative Results in NLP*. May 2025

Hannah Cyberey, Yangfeng Ji, David Evans. [Addressing Both Statistical and Causal Gender Fairness in NLP Models](#). In *Findings of the Association for Computational Linguistics: NAACL*. Jun 2024.

Hannah Cyberey, Yangfeng Ji, David Evans. [Balanced Adversarial Training: Balancing Tradeoffs Between Oversensitivity and Undersensitivity in NLP Models](#). In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Oct 2022.

Hannah Cyberey, Yangfeng Ji, David Evans. [Finding Friends and Flipping Frenemies: Automatic Paraphrase Dataset Augmentation Using Graph Theory](#). In *Findings of the Association for Computational Linguistics: EMNLP*. Nov 2020.

Hannah Cyberey, Yangfeng Ji, David Evans. [Pointwise Paraphrase Appraisal Is Potentially Problematic](#). In *Proceedings of the Association for Computational Linguistics: Student Research Workshop*. Jul 2020.

Preprints & Manuscripts

Marco Gutierrez, Xinyi Leng, **Hannah Cyberey**, Jonathan Schwarz, Ahmed Alaa, Thomas Hartvigsen. [Aligning Language Model Benchmarks with Pairwise Preferences](#). Submitted to *NeurIPS* (under review), May 2026.

Skills

Programming: Python, HTML/CSS, JavaScript

Frameworks/Tools: PyTorch, HuggingFace, Git, Jupyter Notebook, FastAPI, Docker

Libraries: Transformers, Pandas, Scikit-Learn, NumPy, SciPy, Statsmodels, Plotly, Matplotlib, Gradio

Compute Infrastructure: Slurm job scheduling on HPC clusters and multi-GPU servers

Languages: English, Mandarin Chinese

Talks

“Steering Vectors for AI Safety and Auditing,” Invited Talk, Virginia AI Security Initiative, Feb 2026.

Poster: “Steering the CensorShip: Uncovering Representation Vectors for LLM ‘Thought’ Control,” Conference on Language Modeling (COLM), Montréal, Canada, Oct 2025.

Poster: “Addressing Both Statistical and Causal Gender Fairness in NLP Models,” NAACL, Mexico City, Mexico. Jun 2024.

“Debiasing Can Be Complementary,” Invited Talk, AI and Machine Learning Seminar at UVA, Nov 2023.

“Adversarial Attacks and Defenses for NLP Models,” Guest Lecture, UVA CS6501/4501 Data Privacy, Sep 2022.

Teaching Experience

Graduate Teaching Assistant – CS6501/4501 Data Privacy (Fall 2022), CS6501 AI for Social Good (Fall 2021), CS6501 Natural Language Processing (Spring 2021), DS5001 Exploratory Text Analytics (Fall 2020)

Undergraduate Teaching Assistant – Python Programming (Spring 2018)

Leadership & Service

Co-Lead: Causal Learning Reading Group at UVA (Summer 2023)

President: Taiwanese Graduate Student Association at UVA (2022 - 2023)

Reviewer: NLPCC (2021), IJCNLP-AAACL (2023), NeurIPS SoLaR Workshop (2023), NAACL Insights Workshop (2025), ACL Rolling Review (2023-now), COLM (2026), NeurIPS E&D (2026)

Volunteer: IEEE-HKN High School Outreach Program (2021)